List of References for RTG #1: Full Article Titles and Complete Weblinks

1. Thompson K, Reflections on Trusting Trust, *Communications of the ACM*, **27**: 761-763, 1984. https://doi.org/10.1145/358198.358210

2. Emde C, et al., The libRadtran software package for radiative transfer calculations (version 2.0.1), *Geoscientific Models Development*, **9**: 1647-1672, 2016. https://doi.org/10.5194/gmd-9-1647-2016

3. Eriksson P, et al., ARTS, the atmospheric radiative transfer simulator, version 2, *Journal of Quantitative Spectroscopy and Radiative Transfer*, **112**: 1551-1558, 2011. https://doi.org/10.1016/j.jqsrt.2011.03.001

4. Yang J, et al., Advanced radiative transfer modeling system developed for satellite data assimilation and remote sensing applications, *Journal of Quantitative Spectroscopy and Radiative Transfer*, **251**: 107043, 2020. https://doi.org/10.1016/j.jqsrt.2020.107043

5. Dave JV, Meaning of successive iteration of the auxiliary equation in the theory of radiative transfer, *Astrophysical Journal*, **140**: 1292-1303, 1964. https://articles.adsabs.harvard.edu/pdf/1964ApJ...140.1292D

6. Bhartia PK, et al., Solar backscatter UV (SBUV) total ozone and profile algorithm, *Atmospheric Measurement Techniques*, **6**: 2533-2548, 2013. https://doi.org/10.5194/amt-6-2533-2013

7. Herman J, et al., Synoptic ozone, cloud reflectivity, and erythemal irradiance from sunrise to sunset for the whole earth as viewed by the DSCOVR spacecraft from the earth-sun Lagrange 1 orbit, *Atmospheric Measurement Techniques*, **11**: 177-194, 2018. https://doi.org/10.5194/amt-11-177-2018

8. Krotkov NA, et al., The version 3 OMI NO_2 standard product, *Atmospheric Measurement Techniques*, **10**: 3133-3149, 2017. https://doi.org/10.5194/amt-10-3133-2017

9. Lamsal L, et al., Ozone Monitoring Instrument (OMI) Aura nitrogen dioxide standard product version 4.0 with improved surface and cloud treatments, *Atmospheric Measurement Techniques*, **14**: 455-479, 2021. https://doi.org/10.5194/amt-14-455-2021

10. Stamnes K, et al., Numerically stable algorithm for discrete-ordinate-method radiative transfer in multiple scattering and emitting layered media, *Applied Optics*, **27**: 2502-2509, 1988. https://doi.org/10.1364/AO.27.002502

11. Laszlo I, et al., The discrete ordinate algorithm, DISORT for radiative transfer, in Kokhanovsky AA (Ed.), *Light Scattering Reviews: Volume 11*, pp. 3-65, Springer, 2016. https://doi.org/10.1007/978-3-662-49538-4_1

12. Yao B, et al., An accurate and efficient radiative transfer model for simulating all-sky images from Fengyun satellite radiometers, *Science China Earth Sciences*, **63**: 1701-1713, 2020. https://doi.org/10.1007/s11430-020-9617-9

13. Ramon D, et al., Modeling polarized radiative transfer in the ocean-atmosphere system with the GPU-accelerated SMART-G Monte Carlo code, *Journal of Quantitative Spectroscopy and Radiative Transfer*, **222-223**: 89-107, 2019 https://doi.org/10.1016/j.jqsrt.2018.10.017

14. Baes M, et al., Radiative transfer in disc galaxies - III. The observed kinematics of dusty disc galaxies, *Monthly Notices of the Royal Astronomical Society*, **343**: 1081-1094, 2003. https://doi.org/10.1046/j.1365-8711.2003.06770.x

15. Baes M, and Camps P, SKIRT: The design of a suite of input models for Monte Carlo radiative transfer simulations, *Astronomy and Computing*, **12**: 33-44, 2015. https://doi.org/10.1016/j.ascom.2015.05.006

16. Camps P, and Baes M, SKIRT: An advanced dust radiative transfer code with a user-friendly architecture, *Astronomy and Computing*, **9**: 20-33, 2015. https://doi.org/10.1016/j.ascom.2014.10.004

17. Verstocken S, et al., SKIRT: Hybrid parallelization of radiative transfer simulations, *Astronomy and Computing*, **20**: 16-33, 2017. https://doi.org/10.1016/j.ascom.2017.05.003

18. Camps P, and Baes M, SKIRT 9: Redesigning an advanced dust radiative transfer code to allow kinematics, line transfer and polarization by aligned dust grains, *Astronomy and Computing*, **31**: 100381, 2020. https://doi.org/10.1016/j.ascom.2020.100381

19. Merali Z, Computational science: ...Error. *Nature* **467**: 775-777, 2010. https://doi.org/10.1038/467775a

20. Heaton D, and Carver JC, Claims about the use of software engineering practices in science: A systematic literature review, *Information and Software Technology*, **67**: 207-219, 2015. https://doi.org/10.1016/j.infsof.2015.07.011

21. Dubovik O, et al., Polarimetric remote sensing of atmospheric aerosols: Instruments, methodologies, results, and perspectives, *Journal of Quantitative Spectroscopy and Radiative Transfer*, **224**: 474-511, 2019. https://doi.org/10.1016/j.jqsrt.2018.11.024

22. Lenoble J (Ed.), *Radiative Transfer in Scattering and Absorbing Atmospheres: Standard Computational Procedures*, A. Deepak Publishing, 300 pp., 1985.

23. Hansen JE, and Travis LD, Light scattering in planetary atmospheres, *Space Science Reviews*, **16**: 527–610, 1974. https://doi.org/10.1007/BF00168069

24. Hovenier JW, et al., *Transfer of Polarized Light in Planetary Atmospheres: Basic Concepts and Practical Methods*, Kluwer Academic Publishers, 258 pp., 2004.

25. Stamnes K, The theory of multiple scattering of radiation in plane parallel atmospheres, *Reviews of Geophysics*, **24**: 299–310, 1986. https://doi.org/10.1029/RG024i002p00299

26. Chowdhary J, et al., Modeling Atmosphere-Ocean Radiative Transfer: A PACE Mission Perspective, *Frontiers in Earth Science*, **7**: 100, 2019. https://doi.org/10.3389/feart.2019.00100

27. Chandrasekhar S, *Radiative Transfer*, Oxford at the Clarendon Press, 393 pp., 1950.

28. Sobolev VV, *Light Scattering in Planetary Atmospheres*, Pergamon Press, 256 pp., 1976.

29. Kourganoff V, *Basic Methods in Transfer Problems: Radiative Equilibrium and Neutron Diffusion*, Oxford at the Clarendon Press, 281 pp., 1952.

30. Stamnes K, et al., *Radiative Transfer in the Atmosphere and Ocean*, 2nd ed., Cambridge University Press, 528 pp., 2017.

31. Rozanov VV, et al., Radiative transfer through terrestrial atmosphere and ocean: Software package SCIATRAN, *Journal of Quantitative Spectroscopy and Radiative Transfer*, **133**: 13-71, 2014. https://doi.org/10.1016/j.jqsrt.2013.07.004

32. Spurr R, and Christi M, The LIDORT and VLIDORT linearized scalar and vector discrete ordinate radiative transfer models: Updates in the Last 10 Years, in Kokhanovsky AA (Ed.), *Springer Series in Light Scattering: Volume 3*, Springer, pp. 1-62, 2019. https://doi.org/10.1007/978-3-030-03445-0_1

33. Kotchenova SY, et al., Radiative transfer codes for atmospheric correction and aerosol retrieval: intercomparison study, *Applied Optics*, **47**: 2215-2226, 2008. https://doi.org/10.1364/AO.47.002215

34. Hansen JE, and Pollack JB, Near-Infrared Light Scattering by Terrestrial Clouds, *Journal of the Atmospheric Sciences*, **27**: 265-281, 1970. https://doi.org/10.1175/1520-0469(1970)027%3C0265:NILSBT%3E2.0.CO;2

35. van de Hulst HC, *Multiple Light Scattering: Tables, Formulas, and Applications*. Vol. 1, Academic Press, 317 pp., 1980.

36. Rozanov VV, et al., GOMETRAN: A radiative transfer model for the satellite project GOME, the plane-parallel version, *Journal of Geophysical Research Atmospheres*, **102**: 16683-16695, 1997. https://doi.org/10.1029/96JD01535

37. Efremenko DS, et al., Multi-core-CPU and GPU-accelerated radiative transfer models based on the discrete ordinate method, *Computer Physics Communications*, **185**: 3079-3089, 2014. https://doi.org/10.1016/j.cpc.2014.07.018

38. Korkin S, et al., Accuracy of RT code SORD for realistic atmospheric profiles, Proceedings SPIE 10001, Remote Sensing of Clouds and the Atmosphere XXI, 100010B, 2016. https://doi.org/10.1117/12.2241411

39. Korkin SV, et al., Analysis of the radiative transfer equation with highly assymetric phase function, *Journal of Quantitative Spectroscopy and Radiative Transfer*, **112**: 1595-1608, 2011. https://doi.org/10.1016/j.jqsrt.2011.03.016

40. Korkin S, et al., Vector radiative transfer code SORD: Performance analysis and quick start guide, *Journal of Quantitative Spectroscopy and Radiative Transfer*, **200**: 295-310, 2017. https://doi.org/10.1016/j.jqsrt.2017.04.035

41. Korkin S, and Lyapustin A, Matrix exponential in C/C++ version of vector radiative transfer code IPOL, *Journal of Quantitative Spectroscopy and Radiative Transfer*, **227**: 106-110, 2019. https://doi.org/10.1016/j.jqsrt.2019.02.009

42. Lyapustin AI, and Muldashev TZ, Method of spherical harmonics in the radiative transfer problem with non-lambertian surface, *Journal of Quantitative Spectroscopy and Radiative Transfer*, **61**: 545-555, 1999. https://doi.org/10.1016/S0022-4073(98)00041-7

43. Lyapustin A, Radiative transfer code SHARM-3D for radiance simulations over a non-Lambertian nonhomogeneous surface: intercomparison study, *Applied Optics*, **41**: 5607-5615, 2002. https://doi.org/10.1364/AO.41.005607

44. Sayer M, et al., A sea surface reflectance model for (A)ATSR, and application to aerosol retrievals, *Atmospheric Measurement Techniques*, **3**: 813-838, 2010. https://doi.org/10.5194/amt-3-813-2010

45. Sayer M, et al., Extending “Deep Blue” aerosol retrieval coverage to cases of absorbing aerosols above clouds: Sensitivity analysis and first case studies, *Journal*

Geophysical Research Atmospheres, **121**: 4830-4854, 2016. https://doi.org/10.1002/2015JD024729

46. Werdell PJ, and McClain CR, Satellite remote sensing: Ocean color, *Encyclopedia of Ocean Sciences* (3rd ed.), pp. 443-455, 2019. https://pace.oceansciences.org/docs/werdell_and_mcclain_2019_eos3.pdf https://doi.org/10.1016/B978-0-12-409548-9.10817-6

47. Sayer M, et al., Satellite Ocean Aerosol Retrieval (SOAR) Algorithm Extension to S-NPP VIIRS as Part of the "Deep Blue" Aerosol Project, *Journal Geophysical Research Atmospheres*, **123**: 380-400, 2018. https://doi.org/10.1002/2017JD027412

48. Ibrahim A, et al., The relationship between upwelling underwater polarization and attenuation/absorption ratio, *Optics Express*, **20**: 25662-25680, 2012. https://doi.org/10.1364/OE.20.025662

49. Ibrahim A, et al., Retrieval of macro- and micro-physical properties of oceanic hydrosols from polarimetric observations, *Remote Sensing of the Environment*, **186**: 548-566, 2016. https://doi.org/10.1016/j.rse.2016.09.004

50. Ibrahim A, et al., Multiband atmospheric correction algorithm for ocean color retrievals, *Frontiers in Earth Science*, **7**: 1-15, 2019. https://doi.org/10.3389/feart.2019.00116

51. Lyapustin A, et al., MODIS Collection 6 MAIAC algorithm, *Atmospheric Measurement Techniques*, **11**: 5741-5765, 2018. https://doi.org/10.5194/amt-11-5741-2018

52. Lyapustin A, et al., MAIAC Thermal Technique for Smoke Injection Height from MODIS, *IEEE Geoscience and Remote Sensing Letters*, **17**: 730-734, 2020. https://doi.org/10.1109/LGRS.2019.2936332

53. Werdell PJ, et al., The plankton, aerosol, cloud, ocean ecosystem mission: Status, science, advances, Bulletin of the American Meteorological Society, **100**: 1775-1794, 2019. https://doi.org/10.1175/BAMS-D-18-0056.1

54. Werdell PJ, et al., Generalized ocean color inversion model for retrieving marine inherent optical properties, *Applied Optics*, **52**: 2019-2037, 2013, https://doi.org/10.1364/AO.52.002019

55. Press WH, et al., *Numerical Recipes: The Art of Scientific Computing*, Cambridge University Press, 818 pp., 1989.

56. Press WH, et al., *Numerical Recipes: The Art of Scientific Computing*, 3rd ed., Cambridge University Press, 1235 pp., 2007.

57. Arms S, et al., Introducing Students to Scientific Python for Atmospheric Science, *Bulletin of the American Meteorological Society*, **101**: E1492-E1496, 2020. https://doi.org/10.1175/BAMS-D-20-0069.1

58. Wilson RT, Py6S: A Python interface to the 6S radiative transfer model, *Computers & Geosciences*, **51**: 166-171, 2013. https://doi.org/10.1016/j.cageo.2012.08.002

59. Ricchiazzi P, et al., SBDART: A Research and Teaching Software Tool for Plane-Parallel Radiative Transfer in the Earth's Atmosphere, *Bulletin of the American Meteorological Society*, **79**: 2101-2114, 1998. https://doi.org/10.1175/1520-0477(1998)079%3C2101:SARATS%3E2.0.CO;2

60. Ghate VP, et al., The Second ARM Training and Science Application Event: Training the Next Generation of Atmospheric Scientists, *Bulletin of the American Meteorological Society*, **100**: ES5-ES9, 2019. https://doi.org/10.1175/BAMS-D-18-0242.1

61. Lenoble J, et al., A successive order of scattering code for solving the vector equation of transfer in the earth's atmosphere with aerosols, *Journal of Quantitative Spectroscopy and Radiative Transfer*, **107**: 479-507, 2007. https://doi.org/10.1016/j.jqsrt.2007.03.010

62. Hasekamp OP, and Landgraf J, A linearized vector radiative transfer model for atmospheric trace gas retrieval, *Journal of Quantitative Spectroscopy and Radiative Transfer*, **75**: 221-238, 2002. https://doi.org/10.1016/S0022-4073(01)00247-3

63. Lyapustin A, and Knyazikhin Y, Green's function method for the radiative transfer problem. I. Homogeneous non-Lambertian surface, *Applied Optics*, **40**: 3495-3501, 2001. https://doi.org/10.1364/AO.40.003495

64. Evans KF, The spherical harmonics discrete ordinate method for three-dimensional atmospheric radiative transfer, *Journal of the Atmospheric Sciences*, **55**: 429-446, 1998. https://doi.org/10.1175/1520-0469(1998)055<0429:TSHDOM>2.0.CO;2

65. Doicu A, et al., Spectral spherical harmonics discrete ordinate method, *Journal of Quantitative Spectroscopy and Radiative Transfer*, **258**: 107386, 2021. https://doi.org/10.1016/j.jqsrt.2020.107386

66. Zawada D, et al., Systematic comparison of vectorial spherical radiative transfer models in limb scattering geometry, *Atmospheric Measurement Techniques*, **14**: 3953-3972, 2021. https://doi.org/10.5194/amt-14-3953-2021

67. Castellanos P, and da Silva A, A neural network correction to the scalar approximation in radiative transfer, *Journal of Atmospheric and Oceanic Technology*, **36**, 819-832, 2019. https://doi.org/10.1175/JTECH-D-18-0003.1

68. Raissi M, et al., Physics-informed neural networks: A deep learning framework for solving forward and inverse problems involving nonlinear partial differential equations, *Journal of Computational Physics*, **378**: 686-707, 2019. https://doi.org/10.1016/j.jcp.2018.10.045

69. Karniadakis GE, et al., Physics-informed machine learning, *Nature Reviews Physics*, **3**: 422-440, 2021. https://doi.org/10.1038/s42254-021-00314-5

70. Gao M, et al., Inversion of multiangular polarimetric measurements from the ACEPOL campaign: an application of improving aerosol property and hyperspectral ocean color retrievals, *Atmospheric Measurement Techniques*, **13**: 3939-3956, 2020. https://doi.org/10.5194/amt-13-3939-2020

71. Emde C, et al., IPRT polarized radiative transfer model intercomparison project - Phase A, *Journal of Quantitative Spectroscopy and Radiative Transfer*, **164**: 8-36, 2015. https://doi.org/10.1016/j.jqsrt.2015.05.007

72. Herman BM, and Browning SR, A Numerical Solution to the Equation of Radiative Transfer, *Journal of the Atmospheric Sciences*, **22**: 559-566, 1965. https://doi.org/10.1175/1520-0469(1965)022%3C0559:ANSTTE%3E2.0.CO;2

73. Zhai P-W, et al., A vector radiative transfer model for coupled atmosphere and ocean systems based on successive order of scattering method, *Optics Express*, **17**: 2057-2079, 2009. https://doi.org/10.1364/OE.17.002057

74. Herman BM, et al., Comparison of the Gauss–Seidel spherical polarized radiative transfer code with other radiative transfer codes, *Applied Optics*, **34**: 4563-4572, 1995. https://doi.org/10.1364/AO.34.004563

75. Karp AH, et al., Radiative transfer through an arbitrarily thick, scattering atmosphere, *Journal of Quantitative Spectroscopy and Radiative Transfer*, **24**: 391-406, 1980. https://doi.org/10.1016/0022-4073(80)90074-6.

76. Lyapustin AI, Radiative transfer code SHARM for atmospheric and terrestrial applications, *Applied Optics*, **44**: 7764-7772, 2005. https://doi.org/10.1364/AO.44.007764

77. Moler C and Van Loan C, Nineteen Dubious Ways to Compute the Exponential of a Matrix, Twenty-Five Years Later, *SIAM Review*, **45**: 3-49, 2003. https://doi.org/10.1137/S00361445024180

78. Korkin S, et al., Revised and extended benchmark results for Rayleigh scattering of sunlight in spherical atmospheres, *Journal of Quantitative Spectroscopy and Radiative Transfer*, **254**: 107181, 2020. https://doi.org/10.1016/j.jqsrt.2020.107181

79. Krylov VI, *Approximate calculation of integrals*, Dover Publications, 357 pp., 2005.

80. Press, et al., Numerical Recipes in Fortran 90: The Art of Parallel Scientific Computing (vol.2 of Fortran Numerical Recipes), 2nd ed., 1486 pp., 1997.

81. Press WH, et al., Numerical Recipes in C: The Art of Scientific Computing, 2nd ed. Cambridge University Press, 994 pp., 2002.

82. Sykes JB, Approximate integration of the equation of transfer, *Monthly Notices of the Royal Astronomical Society*, **111**: 377-386. https://doi.org/10.1093/mnras/111.4.377

83. Weber HJ and Arfken GB, Essential Mathematical Methods for Physicists, Academic Press, 932 pp., 2003.

84. Blackely RJ, *Potential Theory in Gravity and Magnetic Applications*, Cambridge University Press, 464 pp., 1995.

85. Wauben WMF, et al., Low orders of scattering in a plane-parallel homogeneous atmosphere, *Astronomy and Astrophysics*, **276**: 589-602, 1993 https://ui.adsabs.harvard.edu/abs/1993A&A...276..589W

86. Herman BM, Multiple scatter effects on the radar return from large hail, *Journal of Geophysical Research*, **70**: 1215-1225, 1965. https://doi.org/10.1029/JZ070i005p01215

87. van Diedenhoven, et al., Efficient vector radiative transfer calculations in vertically inhomogeneous cloudy atmospheres, *Applied Optics*, **45**: 5993-6006, 2006. https://doi.org/10.1364/AO.45.005993

88. Herman BM, et al., Numerical technique for solving the radiative transfer equation for a spherical shell atmosphere, *Applied Optics*, **33**: 1760-1770, 1994. https://doi.org/10.1364/AO.33.001760

89. Caudill TR, et al., Evaluation of the pseudo-spherical approximation for backscattered ultraviolet radiances and ozone retrieval, *Journal of Geophysical Research Atmospheres*, **102**: 3881-3890, 1997. https://doi.org/10.1029/96JD03266

90. Loughman R, et al., Gauss-Seidel limb scattering (GSLS) radiative transfer model development in support of the Ozone Mapping and Profiler Suite (OMPS) limb profiler mission, *Atmospheric Chemistry and Physics*, **15**: 3007-3020, 2015. doi: 10.5194/acp-15-3007-2015

91. Eluszkiewicz, et al., A fast code for channel limb radiances with gas absorption and scattering in a spherical atmosphere, *Journal of Quantitative Spectroscopy and Radiative Transfer*, **193**: 31-39, 2017. https://doi.org/10.1016/j.jqsrt.2017.02.010

92. Walter HH, et al., Linearization of a pseudo-spherical vector radiative transfer model, *Journal of Quantitative Spectroscopy and Radiative Transfer*, **85**: 251-283, 2004. https://doi.org/10.1016/S0022-4073(03)00228-0

93. Walter HH and Landgraf J, Towards linearization of atmospheric radiative transfer in spherical geometry, *Journal of Quantitative Spectroscopy and Radiative Transfer*, **95**: 175-200, 2005. https://doi.org/10.1016/j.jqsrt.2004.08.043

94. Ahmad Z and Fraser RS, An iterative radiative transfer code for ocean-atmosphere systems, *Journal of the Atmospheric Sciences*, **39**: 656-665, 1982. https://doi.org/10.1175/1520-0469(1982)039%3C0656:AIRTCF%3E2.0.CO;2

95. Dave JV and Gazdag J, A modified Fourier transform method for multiple scattering calculations in a plane parallel Mie atmosphere, *Applied Optics*, **9**: 1457-1466, 1970. https://doi.org/10.1364/AO.9.001457

96. Mishchenko MI, et al., Errors induced by the neglect of polarization in radiance calculations for rayleigh-scattering atmospheres, *Journal of Quantitative Spectroscopy and Radiative Transfer*, **51**: 491-510, 1994. https://doi.org/10.1016/0022-4073(94)90149-X

97. Chalhoub S and Garcia RDM, The equivalence between two techniques of angular interpolation for the discrete-ordinates method, *Journal of Quantitative Spectroscopy and Radiative Transfer*, **64**: 517-535, 2000. https://doi.org/10.1016/S0022-4073(99)00134-X

98. Zhai P-W, et al., Advanced angular interpolation in the vector radiative transfer for coupled atmosphere and ocean systems, *Journal of Quantitative Spectroscopy and Radiative Transfer*, **115**: 19-27, 2013. https://doi.org/10.1016/j.jqsrt.2012.09.018

99. Kuik F, et al., Benchmark results for single scattering by spheroids, *Journal of Quantitative Spectroscopy and Radiative Transfer*, **47**: 477-489, 1992. https://doi.org/10.1016/0022-4073(92)90107-F

100. Bohren CF and Huffman DR, *Absorption and Scattering of Light by Small Particles*, John Wiley & Sons, 530 pp., 1998.

101. Wiscombe WJ, Improved Mie scattering algorithms, *Applied Optics*, **19**: 1505-1509, 1980. https://doi.org/10.1364/AO.19.001505

102. Mishchenko MI, et al., *Scattering, Absorption, and Emission of Light by Small Particles*, Cambridge University Press, 445 pp., 2002.

103. Mishchenko MI, Comprehensive thematic T-matrix reference database: a 2017-2019 update, *Journal of Quantitative Spectroscopy and Radiative Transfer*, **242**: 106692, 2020. https://doi.org/10.1016/j.jqsrt.2019.106692

104. Dubovik O, et al., Application of spheroid models to account for aerosol particle nonsphericity in remote sensing of desert dust, *Journal of Geophysical Research Atmospheres*, **111**: 1-34, 2006. https://doi.org/10.1029/2005JD006619

105. Oliveira S and Stewart DE, Writing scientific software: A guide to good style. Cambridge University Press, 316 pp., 2006.

106. Gerber R, et al., *The Software Optimization Cookbook*, 2nd ed., Intel Press, 250 pp., 2005. https://dl.acm.org/doi/10.5555/863542

107. Chellappa S, et al., How to Write Fast Numerical Code: A Small Introduction, in Lämmel R. et al. (Eds.), *Generative and Transformational Techniques in Software Engineering II*, *Lecture Notes in Computer Science*, vol. 5235, pp. 196-259, Springer, 2007. https://doi.org/10.1007/978-3-540-88643-3_5

108. Kelly D, et al., Five recommended practices for computational scientists who write software, *Computing in Science & Engineering*, **11**: 48-52, 2009. https://doi.org/10.1109/MCSE.2009.139

109. Wilson G, et al., Best Practices for Scientific Computing, *PLoS Biology*, **12**: e1001745, 2014. https://doi.org/10.1371/journal.pbio.1001745

110. Carver J, et al., Self-perceptions about software engineering: A survey of scientists and engineers, *Computing in Science & Engineering*, **15**: 7-11, 2013. https://doi.org/10.1109/MCSE.2013.12

111. Kernighan BW and Plauger PJ, *The Elements of Programming Style*, 2nd ed, McGraw-Hill, 168 pp., 1978. https://en.wikipedia.org/wiki/The_Elements_of_Programming_Style

112. Dowd K and Severance C, *High Performance Computing*, 2nd ed, O'Reilly Media, 448 pp., 1998.

113. Hager G and Wellein G, *Introduction to High Performance Computing for Scientists and Engineers*, CRC Press, 321 pp., 2010.

114. Buras R, et al., New secondary-scattering correction in DISORT with increased efficiency for forward scattering, *Journal of Quantitative Spectroscopy and Radiative Transfer*, **112**: 2028-2034, 2011. https://doi.org/10.1016/j.jqsrt.2011.03.019

115. Rozanov VV and Lyapustin AI, Similarity of radiative transfer equation: Error analysis of phase function truncation techniques, *Journal of Quantitative Spectroscopy and Radiative Transfer*, **111**: 1964-1979, 2010. https://doi.org/10.1016/j.jqsrt.2010.03.018

116. Kendall R, et al., Development of a weather forecasting code: A case study, *IEEE Software* **25**: 59-65, 2008. https://doi.org/10.1109/MS.2008.86

117. Drake JB, et al., Overview of the Software Design of the Community Climate System Model, *The International Journal of High Performance Computing Applications*, **19**: 177-186, 2005. https://doi.org/10.1177/1094342005056094

118. Easterbrook SM and Johns TC, Engineering the Software for Understanding Climate Change, *Computing in Science & Engineering*, **11**: 65-74, 2009. https://doi.org/10.1109/MCSE.2009.193

119. Clune T and Rood R, Software Testing and Verification in Climate Model Development, *IEEE Software*, **28**: 49-55, 2011. https://doi.org/10.1109/MS.2011.117

120. Pipitone J and Easterbrook S, Assessing climate model software quality: a defect density analysis of three models, *Geoscientific Model Development*, **5**: 1009-1022, 2012. https://doi.org/10.5194/gmd-5-1009-2012

121. Sanders R and Kelly D, Dealing with Risk in Scientific Software Development, *IEEE Software*, **25**: 21-28, 2008. https://doi.org/10.1109/MS.2008.84

122. Brooks P, *The Mythical Man-Month*, Anniversary Ed., Addison-Wesley Longman Publishing, 322 pp., 1995. https://en.wikipedia.org/wiki/The_Mythical_Man-Month

123. Prechelt L, Empirical comparison of seven programming languages, *Computer*, **33**: 23-29, 2000. https://doi.org/10.1109/2.876288